

DESIGNING FOR INTENTIONAL AI USE

Adaptive Response Depth, Reflective Friction, and Privacy-Preserving Behavioral Signals in Generative-AI Interfaces

Research-backed product case study

Human-AI Interaction · Cognitive Security · AI Product Design · Privacy

Research question: How can generative-AI interfaces adapt to the way people actually consume AI-generated information while preserving cognitive agency, privacy, and user control?

Project premise. Generative AI has dramatically reduced the cost of producing information. But generation and consumption are not equivalent. Current interfaces can produce thousands of words without knowing whether a user carefully read them, searched for one fact, skimmed most of the response, or immediately moved on.

Important boundary. The proposed system does not claim to detect learning, attention, intelligence, or comprehension from scrolling. Behavioral telemetry is treated as noisy evidence about interaction style, and cognitive outcomes remain questions for empirical evaluation.

Prepared as a portfolio research and product-design case study · September 2026

Abstract

Generative AI systems can produce detailed explanations, analyses, summaries, and recommendations within seconds. However, generation and consumption are not equivalent: an AI system may produce a lengthy response without knowing whether the user carefully read it, searched it for one fact, skimmed most of it, or immediately moved on. Research on generative AI presents a more complicated picture than the common claim that AI simply makes users think less. Generative AI has been associated with improvements in learning, cognitive engagement, motivation, and task performance under some conditions. At the same time, studies identify risks involving overreliance, cognitive offloading, reduced critical-thinking effort, weaker long-term retention, and miscalibrated confidence. Emerging work suggests that interface and instructional design can influence these outcomes. This case study proposes a privacy-preserving adaptive layer for generative-AI interfaces that uses behavioral interaction signals - including response coverage, dwell time, scrolling patterns, revisitation, selection, and follow-up timing - to estimate how users tend to interact with generated responses. These signals are treated as interaction proxies rather than measurements of comprehension. The system adapts presentation depth, uses progressive disclosure, and introduces optional reflective friction when repeated behavior suggests that a different interaction style may be useful. The architecture separates task complexity from preferred response depth and prioritizes local processing of behavioral telemetry. A browser/API proof of concept is proposed, followed by an empirical evaluation plan separating usability, engagement, comprehension, retention, calibration, and privacy acceptance.

Keywords: generative AI; human-AI interaction; cognitive offloading; metacognition; adaptive interfaces; progressive disclosure; privacy; reflective friction

1. Introduction

Generative AI has significantly reduced the cost of producing information. A user can request an explanation, literature review, debugging analysis, itinerary, study guide, or strategic recommendation and receive hundreds or thousands of words almost immediately.

Yet the interface generally knows very little about what happens next. A 1,500-word answer may be read carefully. It may also be scrolled through in seconds. The user might repeatedly return to one paragraph, copy a particular sentence, inspect references, or immediately submit another prompt without meaningfully interacting with the previous response.

From the system's perspective, these behaviors often collapse into approximately the same sequence: **Prompt -> Response -> Next Prompt**. This case study begins from the premise that this is an important missing interaction layer.

The relevant question is not simply whether AI produces a correct answer. It is also whether the **form, depth, timing, and cognitive demands** of that answer match what the user is trying to accomplish. The design opportunity is therefore not to maximize generated content, nor to minimize it indiscriminately, but to better align generation with human attention, intent, and engagement.

2. Research Questions

RQ1 - Behavioral adaptation. Can observable interaction signals help a generative-AI interface estimate an appropriate level of response depth without incorrectly treating those signals as measurements of comprehension?

RQ2 - Reflective interaction. Can contextual, optional moments of friction encourage users to reflect on AI-generated material when reflection is useful, without making routine AI interaction unnecessarily burdensome?

RQ3 - Resource efficiency. Can response-depth adaptation reduce unnecessary output-token generation while preserving task quality and access to deeper information when users need it?

RQ4 - Privacy. Can such personalization be achieved without building a detailed centralized record of what users read, write, ask, or ignore?

3. Why This Problem Matters

3.1 Generative AI changes cognitive effort rather than simply eliminating it

A CHI 2025 study of 319 knowledge workers collected 936 examples of real-world generative-AI use. Participants reported critical thinking in roughly 60% of the tasks studied. Higher confidence in AI was associated with less reported critical-thinking effort, while greater confidence in one's own abilities predicted greater critical engagement. Importantly, AI did not simply remove thinking: users described shifts from information gathering toward verification, from direct problem solving toward integrating AI-generated responses, and from task execution toward task stewardship (Lee et al., 2025).

Design implication: the objective should not be to make users do more work. It should be to preserve useful human judgment where human engagement matters.

4. Cognitive Offloading: Useful Tool or Cognitive Crutch?

Cognitive offloading refers to shifting cognitive work onto an external resource. Humans have always done this: notebooks offload memory, calculators offload arithmetic, maps offload spatial recall, and search engines offload information retrieval. Generative AI expands the range and complexity of work that can be delegated.

A recent randomized controlled trial involving 120 undergraduate students compared ChatGPT-supported study with traditional study methods. On a surprise retention test 45 days later, the ChatGPT-assisted group scored lower than the traditional-study group. The authors interpret this as evidence that unrestricted AI assistance can reduce cognitive effort necessary for durable learning (Barcaui, 2025).

Other recent work similarly identifies a trade-off between immediate performance and later skill development under high levels of AI-supported cognitive offloading. These findings make cognitive offloading important to study, but they do not justify treating all offloading as harmful. Efficient delegation can be useful; the design concern is whether important cognitive processes are consistently delegated without awareness or choice.

5. The Evidence Is Not 'AI Makes People Think Less'

A responsible design case must also examine evidence in the opposite direction. Recent meta-analyses report positive effects of generative AI on academic performance, higher-order thinking, motivation, and several forms of engagement, with substantial variation across instructional strategy and context. Metacognition itself has not shown uniformly positive effects. The literature therefore points toward a conditional relationship: outcomes depend strongly on how AI is incorporated into a task and how the interaction is designed.

Oversimplified framing	More defensible framing
AI automatically weakens thinking	Outcomes depend on interaction design, task, context, and user behavior.
More AI use = less learning	Some AI-supported learning produces meaningful benefits; some forms of offloading may reduce durable learning.
Cognitive offloading is inherently bad	Offloading can increase efficiency but may be harmful when important cognitive processes are consistently delegated.
Longer engagement means better learning	Behavioral engagement and cognitive engagement are related but distinct.
AI should make users work harder	AI should preserve meaningful cognitive activity while removing unnecessary work.

6. Metacognition as the Missing Layer

Metacognition concerns awareness and regulation of one's own thinking: recognizing what one knows, identifying uncertainty, selecting strategies, evaluating performance, and deciding when additional effort is necessary. CHI 2024 work on the metacognitive demands and opportunities of generative AI argues that systems can support self-evaluation through context-sensitive prompts that encourage users to reflect on goals, strategies, knowledge, and confidence (Tankelevitch et al., 2024).

A 2026 study of 141 university learners comparing ordinary GenAI-assisted writing feedback with structured metacognitive instruction reported improvements in strategy knowledge, metacognitive strategy use, uptake of higher-level feedback, and revisions addressing content and organization. This suggests that the interface around AI-generated information can matter as much as the information itself.

7. Designed Friction

Digital products traditionally treat friction as something to eliminate. For many workflows this is appropriate. Human-AI interaction introduces situations in which a small amount of deliberate friction may instead protect human judgment.

Buçinca, Malaya, and Gajos experimentally studied cognitive forcing functions designed to make users engage more deliberately with AI-supported decisions. These interventions reduced overreliance on incorrect AI recommendations compared with conventional explanation interfaces, although participants generally preferred the less demanding interfaces. More recent work continues to investigate cognitive forcing functions and question-based interaction as mechanisms for reducing blind reliance.

Design principle: friction should be selective, contextual, reversible, and proportionate to the cognitive stakes of the task.

8. Progressive Disclosure

One possible solution to excessive AI output is not simply shorter answers, but progressive disclosure. Instead of immediately presenting every available detail, an AI system can provide a compact useful response and make additional depth easily accessible. Recent experimental work on exploratory search with generative AI suggests that progressive disclosure can increase engagement and knowledge change while reducing unnecessary cognitive demands.

TRADITIONAL AI	PROGRESSIVE AI
Prompt	Prompt
v	v
Generate everything	Give useful core
v	v
User searches answer	Expose optional depth
v	
User chooses depth	

Figure 1. Two different information-delivery models.

9. Scrolling Is Not Thinking

The project's initial intuition came from a recognizable behavior: a system generates a large response, the user rapidly scrolls through most of it, and the user immediately asks another question. It is tempting to conclude that the user did not read or understand the answer. That conclusion would be scientifically irresponsible.

Scrolling is behavior. Comprehension is cognition. The two are not equivalent.

Digital reading research demonstrates why this distinction matters. Scrolling, scanning, searching, revisiting, and deep reading can all be legitimate strategies depending on the task. Mouse movement and interaction traces can provide useful signals about interaction patterns, but they remain proxies rather than direct observations of cognition. The proposed system therefore infers interaction tendencies, not learning or intelligence.

10. Product Concept

The proposed system introduces an adaptive layer between a generative-AI response and the person consuming it. Its purpose is to learn how a user prefers to interact with generated information while preserving the user's ability to override every adaptation.

```
USER REQUEST
|
v
TASK CONTEXT
|
v
AI RESPONSE
|
v
INTERACTION SIGNALS
response coverage / dwell / scrolling
revisitation / selection / follow-up timing
|
v
LOCAL BEHAVIOR MODEL
|
v
ADAPTATION
response depth / disclosure / optional reflection
```

Figure 2. Proposed adaptive interaction loop.

11. What the System Can Observe

Signal	Possible interpretation	Important alternative explanation
Response coverage	How much of the response entered the viewport	Visibility does not imply reading
Dwell time	Sustained presence around content	User may be distracted or inactive
Scroll velocity	Rapid navigation versus slower movement	User may already know the material
Revisitation	Returning to earlier content	Could indicate interest, confusion, or verification
Text selection	Focus on a specific passage	Could simply precede copying
Copy behavior	Possible downstream use	Does not indicate understanding
Time to next prompt	Immediate continuation versus extended interaction	User may be doing work elsewhere
Expansion behavior	Requests for additional detail	May indicate interest or insufficient original response
Citation interaction	Verification-oriented behavior	Absence does not imply blind trust

No individual signal is treated as meaningful enough to trigger strong adaptation. The system relies on patterns over repeated interactions.

12. Interaction States, Not Cognitive Diagnoses

The system should not categorize a person as learning, lazy, dependent, or inattentive. Instead, it can identify interaction patterns.

```
Rapid traversal + short dwell + little revisitation
-> RAPID RESPONSE NAVIGATION

Longer dwell + backward scrolling + selections
-> DELIBERATE RESPONSE EXPLORATION

Fast navigation + keyword search + selected passage
-> TARGETED INFORMATION RETRIEVAL
```

Figure 3. Interaction patterns are descriptive rather than diagnostic.

None of these patterns is inherently better. A person looking for a command-line flag should not be forced to deeply read an 800-word explanation.

13. Separate Response Depth From Model Capability

Rapid scrolling should not automatically cause the system to select a weaker model. Response length and reasoning capability are different variables. A difficult cryptographic debugging request may require strong reasoning but only a four-sentence answer.

```
REQUEST
|
+-----+-----+
| |
v v
TASK COMPLEXITY PRESENTATION NEED
| |
v v
model / reasoning response depth
allocation and structure
| |
+-----+-----+
|
v
RESPONSE
```

Figure 4. Reasoning allocation and presentation depth are independent design dimensions.

Current model APIs increasingly support this distinction by exposing controls for model choice, reasoning effort, thinking levels, and output limits. The prototype should therefore optimize presentation separately from capability.

14. Adaptive Response Depth

The system would begin conservatively. A new user receives ordinary responses. Over time, interaction evidence might reveal that long answers are routinely traversed quickly while shorter responses are examined more carefully. Rather than silently changing behavior, the system asks whether the user would like future responses to start with the essential answer.

```
CORE ANSWER

The immediate issue is nonce reuse.
Generate a unique nonce for every encryption operation.

[ why this matters ]
[ Show the vulnerable flow ]
[ Walk me through the cryptography ]
[ Show corrected code ]
```

Figure 5. Progressive depth preserves access to detail while reducing initial information load.

15. Reflective Friction

Adaptation addresses inefficient presentation; it does not address passive dependency. A second layer introduces optional reflective friction. If a user is in a learning-oriented session and repeatedly requests explanations while rapidly moving through each response, the system might offer a choice rather than issuing a warning.

You've been moving quickly through the last few explanations.

Give me a shorter explanation · Quiz me on what we've covered · Let me explain it back · Continue normally

The interaction creates an opportunity for reflection without asserting that the user behaved incorrectly.

16. User Intent Must Matter

Context	Example	Appropriate response
Fact retrieval	What port does PostgreSQL use by default?	Minimize friction; concise answer.
Learning	Teach me process scheduling for an exam.	Offer progressive disclosure or reflection after repeated patterns.
Production work	Review these logs and identify the failure.	Concise recommendation with expandable evidence.

A privacy-preserving implementation could provide lightweight explicit session modes such as **Quick answer**, **Learn**, **Work through something**, and **Explore**. Explicit modes reduce the need to infer intent from behavior.

:

17. When Should the System Intervene?

```
Interaction pattern detected
|
v
Is it repeated?
/ \
no yes
| |
v v
observe consider context
|
+-----+-----+
| |
v v
retrieval/work learning/explore
| |
v v
adapt depth offer reflection
```

Figure 6. Intervention requires repeated evidence and context.

No intervention from one scroll event.

No claim that behavior demonstrates comprehension.

No friction where speed is the user's explicit goal.

Every reflective intervention must be dismissible.

18. From Friction to Agency

Research on cognitive forcing functions suggests that stronger cognitive interventions can reduce overreliance while sometimes reducing user preference. A system that constantly forces reflection may simply teach users to dismiss reflection prompts. The objective is therefore not maximum reflection, but appropriate reflection.

Useful friction = potential cognitive benefit - interruption cost
This is a design heuristic, not a validated mathematical model.

19. Privacy Creates the Central Product Tension

The system's personalization requires observing behavior. That introduces a privacy contradiction: to build an interface that respects human attention, the system must first observe attention-related behavior. A naive implementation could collect prompts, responses, scroll history, time spent reading, copied passages, and behavior across sessions. Such a profile is unnecessary.

20. Local-First Architecture

```
DEVICE

AI response rendered
|
v
interaction events
|
+-----+-----+
| | |
scroll dwell revisit
| | |
+-----+-----+
|
v
LOCAL FEATURE ENGINE
|
v
aggregated statistics
|
+-----+-----+
| |
discard raw retain local
events preferences
```

Figure 7. Local-first behavioral processing.

The system should preferentially store abstract states such as response-depth preference, progressive-disclosure preference, reflection-prompt frequency, and confidence in those preferences rather than retaining detailed event histories.

21. Data Minimization

Data	Default handling
Prompt content	Not stored by extension
AI response text	Not stored
Full browser history	Never collected
Exact copied text	Not stored
Raw cursor trajectory	Process transiently, then discard
Raw scroll events	Process transiently, then discard
Aggregated response-coverage metrics	Local
Response-depth preference	Local
Intervention preference	Local
Session intent	Session-local unless user chooses otherwise

22. Browser-Extension Privacy Requirements

Chrome extensions can execute content scripts on specified pages and observe page-level interaction when appropriate permissions are granted. However, extensions should request the narrowest permissions necessary. Chrome Web Store policy also restricts collection of browsing activity to data necessary for prominently disclosed user-facing features and emphasizes limiting data collection to the extension's stated purpose. A local-first architecture therefore aligns both with privacy principles and with platform expectations.

23. Platform Feasibility

The initial idea imagined an extension operating across ChatGPT, Claude, Gemini, and other AI systems. Technically, this should be divided into two layers: an **interaction companion** that can observe and adapt presentation at the browser level, and an **adaptive generation engine** that changes generation parameters through official APIs.

Deeply automating consumer AI websites would create brittle integrations and potential terms-of-service problems. A more robust prototype should use an official model API for generation while demonstrating browser-level behavioral sensing separately.

24. Current Platform Assessment

Platform	Browser sensing	Consumer-site model control	Official API
ChatGPT	Technically possible with page permissions	Not recommended as core architecture	Yes
Claude	Technically possible	Not recommended as core architecture	Yes
Gemini	Technically possible	Not recommended as core architecture	Yes, subject to current terms

Platform terms and API capabilities change frequently. Before distribution, the implementation should be re-reviewed against current consumer terms, developer terms, browser-store policies, and API data-use requirements.

25. Recommended Prototype Architecture

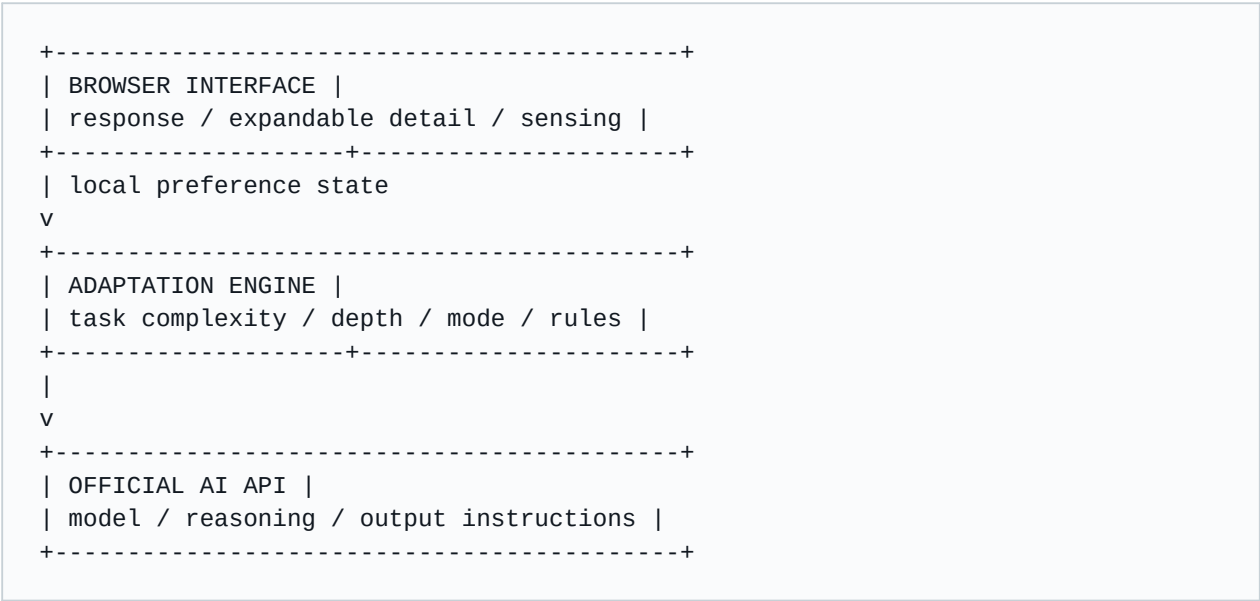


Figure 8. Recommended proof-of-concept architecture.

26. MVP Scope

Component	MVP implementation
Interaction sensing	Scroll, viewport coverage, dwell, revisitation, selection
Local profile	Aggregate behavioral features stored in browser storage
Adaptive response	Concise / balanced / detailed generation
Reflective support	Optional intervention after predefined repeated patterns

The first engagement estimator should be rule-based. That is preferable to claiming a sophisticated classifier before collecting training data. If a long response is traversed rapidly, effective dwell is short, and the pattern repeats, the system can offer to start future responses more concisely. It still does not claim the response was unread.

27. Why We Should Not Build an 'Attention Score'

A single score such as 'Engagement: 43%' would imply measurement precision the system does not possess. It could also invite incorrect equivalences between engagement, attention, and comprehension. The user-facing system should instead expose understandable observations such as: *You usually expand detailed explanations rather than reading them upfront, or Long responses are frequently traversed quickly. Start shorter?*

28. Proposed Behavioral Model



Figure 9. Multiple interpretable tendencies are preferable to a pseudo-precise engagement score.

29. Reflection Without Nagging

Level	Behavior
0 - No intervention	Normal interaction
1 - Presentation adaptation	Offer to start with the key answer
2 - User-controlled expansion	Explain why / show example / go deeper
3 - Reflection opportunity	Offer to check understanding before continuing
4 - Active-learning mode	Explicitly chosen mode using retrieval practice or teach-back

The product never silently locks information behind a test.

30. Different Tasks Need Different Friction

Task	Appropriate system behavior
Quick factual lookup	Minimize friction
Troubleshooting	Concise diagnosis + expandable reasoning
Research exploration	Rich explanation + source exploration
Studying	Reflection options + retrieval practice
Writing	Encourage user contribution before full generation when desired
Brainstorming	Preserve divergent exploration
High-stakes decision	Encourage verification rather than simply shortening output

31. Token Efficiency

The original idea also contains an efficiency question. If a system repeatedly generates extensive visible output that a user consistently chooses not to explore, progressive disclosure could defer generation until deeper content is requested. This does not mean every skipped paragraph represents wasted computation. It creates an experimentally testable hypothesis: can adaptive response depth reduce generated output tokens without reducing perceived usefulness, comprehension, or task success?

32. Design Hypotheses

H1 - Adaptive depth. Users receiving behavior-informed response depth will generate fewer unnecessary visible output tokens than users receiving static responses without significantly reducing task success.

H2 - Progressive disclosure. Progressive disclosure will produce greater targeted interaction with detailed content than presenting equivalent detail simultaneously.

H3 - Reflective friction. Optional context-sensitive reflection prompts will improve verification, recall, or calibration during learning-oriented tasks compared with equivalent AI assistance without reflection prompts.

H4 - Privacy-preserving personalization. Users will report greater comfort with personalization when raw behavioral traces remain on-device and the system exposes which preferences are being inferred.

These are proposed hypotheses, not findings.

33. Evaluation Plan

33.1 Phase I - Formative research

Semi-structured interviews would explore how people recognize when AI is giving too much information; why they skim; when long answers are valued; whether they perceive AI use as passive or active; what interventions feel helpful versus patronizing; and what behavioral monitoring they consider acceptable.

33.2 Phase II - Controlled interaction study

Participants would complete retrieval, learning, and applied-reasoning tasks. Conditions could compare a standard generated response, an adaptive interface using behavior-informed depth and progressive disclosure, and an adaptive interface with contextual reflection prompts.

Condition	Interface
Baseline	Standard generated response
Adaptive	Behavior-informed depth + progressive disclosure
Reflective	Adaptive interface + contextual reflection prompts

34. Evaluation Metrics

Construct	Possible measurement
Task success	Accuracy / rubric score
Comprehension	Post-task questions
Retention	Delayed recall
Calibration	Confidence compared with actual accuracy
Reliance	Acceptance of deliberately incorrect AI suggestions
Engagement behavior	Dwell, revisitation, expansion
Efficiency	Time + generated tokens
User agency	Perceived control scale
Intervention cost	Dismissals / annoyance / interruption rating
Privacy acceptance	Perceived intrusiveness and willingness to enable telemetry

Behavior alone should not serve as the outcome measure. This prevents circular reasoning in which interaction is simultaneously treated as both the intervention signal and proof of success.

35. What Would Falsify the Product Idea?

A serious case study should describe evidence that could tell us the product is unnecessary. The concept should be reconsidered if users accurately manage desired response length themselves with negligible effort; behavioral signals add little predictive information beyond explicit preferences; adaptive responses reduce task performance; reflection prompts create substantially more annoyance than cognitive benefit; progressive disclosure causes users to miss important information; or privacy concerns make behavioral adaptation unacceptable. A negative result would still be valuable HAI research.

36. Risks and Failure Modes

Risk	Design response
Misinterpreting behavior	Infer presentation preference, not cognitive ability.
Excessive paternalism	Interventions are invitations, not restrictions.
Automation of metacognition	Expose choices; let users initiate deeper modes.
Privacy creep	Establish strict architectural data boundaries from the beginning.
Platform fragility	Do not make consumer-site DOM manipulation the core dependency.
Incorrect model downgrading	Keep reasoning allocation independent from presentation depth.

37. Ethical Position

The system is not designed to maximize engagement. Many digital systems optimize engagement because more interaction benefits the platform. This project's objective is different: **the user should spend exactly as much attention as their goal requires**. Sometimes that means slowing down. Sometimes it means getting an answer in ten seconds and leaving. The system should not decide which is morally preferable.

38. Design Principles

1. Behavior is evidence, not diagnosis.
2. User intent outranks behavioral inference.
3. Response depth and model capability are separate variables.

4. Friction must be contextual and optional.
5. Personalization should reveal itself rather than operate invisibly.
6. Raw behavioral data should remain local whenever technically possible.

39. What the Prototype Demonstrates

This project is deliberately positioned between research and engineering. The prototype does not need to prove that it improves cognition. It needs to demonstrate that the proposed interaction model is technically plausible. The research contribution is the reasoning connecting AI capability, cognitive offloading, human engagement, observable behavior, privacy-preserving inference, adaptive interfaces, and reflective friction.

```
AI CAPABILITY
|
v
COGNITIVE OFFLOADING
|
v
HUMAN ENGAGEMENT
|
v
OBSERVABLE BEHAVIOR
|
v
PRIVACY-PRESERVING INFERENCE
|
v
ADAPTIVE INTERFACE
|
v
REFLECTIVE FRICTION
```

Figure 10. Conceptual chain underlying the product case study.

40. Implementation Complexity

Component	Complexity	Recommendation
Literature-backed case study	Medium	Build now
Interaction-tracking extension	Medium	Build
Local preference model	Low-Medium	Build
Adaptive response UI	Medium	Build
Progressive disclosure	Low-Medium	Build
Reflection interventions	Low	Build
API-connected AI prototype	Medium	Build
Automatic cross-platform model control	High / brittle	Do not make core MVP
Accurate learning detection	Research problem	Do not claim
Production behavioral ML classifier	High	Future work
Longitudinal cognitive-impact validation	High	Future research

41. Positioning the Project

The project should not be described as an extension that detects whether people actually read ChatGPT, an AI system that prevents cognitive decline, or a system that automatically gives 'lazy' users shorter answers.

Academically defensible positioning: A research-backed exploration of how behavioral interaction signals, progressive disclosure, and reflective friction could help generative-AI interfaces better align response depth with user intent while preserving privacy and cognitive agency.

42. Portfolio Contribution

Area	What the project demonstrates
Human-AI Interaction	How people consume, interpret, and respond to AI-generated information
Psychology	Metacognition, cognitive offloading, critical thinking, reliance, and attention
Product Design	Progressive disclosure, personalization, interruption design, and user control
Privacy	Local behavioral processing, minimization, permission boundaries, and transparency
AI Systems	Reasoning effort, model routing, generation controls, and API architecture

43. Final Product Thesis

Generative AI has dramatically improved its ability to produce information. Interfaces have not improved at the same rate in understanding how humans want to consume that information. The resulting design problem is not simply that answers are too long, nor that users scroll too quickly. It is that current AI interaction largely treats **generation as the endpoint**.

Human cognition begins after generation. A more human-centered system would recognize that people use AI for fundamentally different purposes: sometimes to retrieve, sometimes to learn, sometimes to explore, sometimes to delegate, and sometimes to think alongside another system. The objective should therefore not be to maximize either AI assistance or human effort. It should be to allocate both deliberately.

The future of useful AI may depend not only on how well models know when to think more, but also on how well their interfaces know when humans want to think more themselves.

References

- [1] Lee, H.-P., Sarkar, A., Tankelevitch, L., Drosos, I., Rintel, S., Banks, R., & Wilson, N. (2025). The Impact of Generative AI on Critical Thinking: Self-Reported Reductions in Cognitive Effort and Confidence Effects From a Survey of Knowledge Workers. *CHI 2025*. DOI: 10.1145/3706598.3713778.
<https://doi.org/10.1145/3706598.3713778>
- [2] Tankelevitch, L., et al. (2024). The Metacognitive Demands and Opportunities of Generative AI. *CHI 2024*. DOI: 10.1145/3613904.3642902.
<https://doi.org/10.1145/3613904.3642902>
- [3] Klingbeil, A., Grützner, C., & Schreck, P. (2024). Trust and Reliance on AI - An Experimental Study on the Extent and Costs of Overreliance on AI. *Computers in Human Behavior*, 160, 108352.
<https://doi.org/10.1016/j.chb.2024.108352>
- [4] Buçinca, Z., Malaya, M. B., & Gajos, K. Z. (2021). To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-Assisted Decision-Making. *Proceedings of the ACM on Human-Computer Interaction*. DOI: 10.1145/3449287.
<https://doi.org/10.1145/3449287>
- [5] Danry, V., Pataranutaporn, P., Mao, Y., & Maes, P. (2023). Don't Just Tell Me, Ask Me: AI Systems that Intelligently Frame Explanations as Questions Improve Human Logical Discernment Accuracy. *CHI 2023*. DOI: 10.1145/3544548.3580672.
<https://doi.org/10.1145/3544548.3580672>
- [6] Chen, S., & Cheung, A. C. K. (2025). Effect of Generative Artificial Intelligence on University Students' Learning Outcomes: A Systematic Review and Meta-Analysis. *Educational Research Review*.
<https://www.sciencedirect.com/science/article/pii/S1747938X25000740>
- [7] Xia, Q., Li, W., Yang, Y., Weng, X., & Chiu, T. K. F. (2025). A Systematic Review and Meta-Analysis of the Effectiveness of Generative Artificial Intelligence on Students' Motivation and Engagement. *Computers and Education: Artificial Intelligence*, 9, 100455.
<https://doi.org/10.1016/j.caeai.2025.100455>
- [8] Heung, Y. M. E., & Chiu, T. K. F. (2025). How ChatGPT Impacts Student Engagement From a Systematic Review and Meta-Analysis Study. *Computers and Education: Artificial Intelligence*, 8, 100361.
<https://doi.org/10.1016/j.caeai.2025.100361>
- [9] Barcaui, A. (2025). ChatGPT as a Cognitive Crutch: Evidence From a Randomized Controlled Trial on Knowledge Retention. *Social Sciences & Humanities Open*, 12, 102287.
<https://www.sciencedirect.com/science/article/pii/S2590291125010186>
- [10] Yan, D. A. (2026). Metacognitive Instruction to Enhance EFL Writers' Metacognition, Feedback Uptake, and Revision Performance in a GenAI-Assisted Feedback Context. *International Journal of Applied Linguistics*. DOI: 10.1111/ijal.70328.
<https://doi.org/10.1111/ijal.70328>
- [11] Krenca, K., Taylor, E., & Deacon, S. H. (2024). Scrolling and Hyperlinks: The Effects of Two Prevalent Digital Features on Children's Digital Reading Comprehension. *Journal of Research in Reading*, 47, 269-291.
<https://doi.org/10.1111/1467-9817.12468>
- [12] Kim, N., & Ji, Y. G. (2026). Exploratory Search With Generative AI: An Empirical Study on the Impact of Interaction Design Strategies on Information Exploration and Cognitive Load. *International Journal of Human-Computer Studies*, 210, 103771.
<https://www.sciencedirect.com/science/article/abs/pii/S1071581926000467>
- [13] Ghosh, A., Sarkar, A., Lindley, S., & Poelitz, C. (2026). An Experimental Comparison of Cognitive Forcing Functions for Execution Plans in AI-Assisted Writing. *Microsoft Research preprint*.
<https://www.microsoft.com/en-us/research/publication/an-experimental-comparison-of-cognitive-forcing-functions-for-execution-plans-in-ai-assisted-writing-effects-on-trust-overreliance-and-perceived-critical-thinking/>
- [14] Google Chrome Developers. Chrome Extension Permissions and Limited Use Policies. Accessed September 2026.
<https://developer.chrome.com/docs/webstore/program-policies/limited-use>

[15] OpenAI. Model Selection; Reasoning Models; Terms of Use. Accessed September 2026.
<https://developers.openai.com/api/docs/guides/model-selection>

[16] Anthropic. Effort Controls for Claude; API Access for Individual Developers. Accessed September 2026.
<https://platform.claude.com/docs/en/build-with-claude/effort>

[17] Google AI for Developers. Gemini Thinking Controls and Gemini API Additional Terms of Service. Accessed September 2026.
<https://ai.google.dev/gemini-api/docs/thinking>

Research note. This document distinguishes published findings from proposed product hypotheses. The behavioral model, intervention rules, and evaluation design are research proposals and should not be represented as validated outcomes until empirically tested.